MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BBA (HONOURS) REGULAR EXAMINATION OCTOBER 2025

THIRD SEMESTER

Core Course(CC) - MG3CCRBBA202 – BUSINESS RESEARCH METHODOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 70

Part A

Short Answer Questions(four to five sentences)

1a. Explain survey research with an example of its application. [U] / [CO1]

The survey is the most commonly used social research method for collecting data from individuals in a population of interest. Survey research is a **method of collecting information from a large group of people** by using questionnaires, interviews, or online forms. It helps researchers gather data on people's opinions, attitudes, behaviors, or experiences related to a particular topic. **2 marks**

This approach is widely used in social sciences, public health, market research, and politics to understand opinions, behaviors, and characteristics. Any valid example – **1 mark.**

1b. Name one ancient scholar and his contribution to early research methods. [U] / [CO1]

- Plato (427–347 BCE) – Emphasized reasoning and logical thinking as a way to discover truth; introduced the concept of ideal forms and deductive reasoning.
- Socrates (470–399 BCE) – Developed the Socratic method, a systematic questioning technique to stimulate critical thinking and uncover underlying beliefs.
- Hippocrates (460–370 BCE) – Known as the *Father of Medicine*; promoted observation and documentation in medical research instead of relying on superstition.
- Pythagoras (c. 570–495 BCE) – Used mathematical reasoning and introduced early forms of hypothesis testing in geometry and number theory.
- Thales of Miletus (c. 624–546 BCE) – Among the first to apply rational inquiry and observation to explain natural phenomena rather than attributing them to myths.
- Confucius (551–479 BCE) – In China, emphasized systematic study and reflection on human behavior and ethics as a way of gaining knowledge.
- Chanakya (Kautilya) (c. 350–283 BCE) – In India, authored the *Arthashastra*, which demonstrates systematic data collection and analysis in governance and economics.

2a. Mention any three reasons why selecting a familiar topic is advisable for a researcher. [U] / [CO2]

- **Better Understanding:**
  A familiar topic allows the researcher to start with a strong background knowledge, making it easier to understand concepts, theories, and issues related to the study.
- **Easy Access to Information:**
  Since the researcher already knows the field, it becomes easier to identify reliable sources, collect relevant data, and interpret the findings accurately.
- **Saves Time and Effort:**
  Working on a familiar topic reduces the time needed for preliminary learning or exploration, allowing the researcher to focus more on analysis and writing.

Provide marks for any other valid points.

2b. In what way does hypothesis testing reduce research bias and improve decision-making? [U] / [CO2]

Reduce bias - Prevents researchers from relying on personal opinions or assumptions

Improve decision-making - By testing whether a hypothesis is supported or rejected through data analysis, researchers can make logical and evidence-based conclusions, leading to more accurate and reliable results.

3a. Outline the basic steps in the procedure of conducting a literature review. [U] / [CO2]

Define the Research Question/Problem, Search for Literature, Evaluate and Select Sources, Organize the Literature, Analyze and Synthesize, Write the Review, Cite Sources Properly.

3b. Explain the technique involved in defining a research problem [U] / [CO2]

Statement of the problem in a general way; Understanding the nature of the problem; Surveying the available literature ; Developing the ideas through discussions; and Rephrasing the research problem into a working proposition.

4a. A researcher wants to measure students' stress levels using a questionnaire. Suggest two ways to ensure the reliability of the instrument. [A] / [CO3]

Reliability can be improved through pilot testing, test–retest method, clear wording, and internal consistency checks.

4b. What is Multi stage sampling? [U] / [CO3]

Multi-stage sampling is a probability sampling technique in which the researcher selects samples through two or more stages, instead of drawing the final sample directly from the entire population. It is especially useful when the population is large and spread across wide areas, making it difficult and costly to list or contact every individual.

5a. What is bivariate analysis? Give one simple business-related example. [A] / [CO4]

Bivariate analysis is a statistical technique used to study the relationship between two variables. It helps researchers understand how one variable changes in response to another. A company wants to study the relationship between advertising expenditure and sales revenue.

5b. 1. Two groups in your class used different coding systems for the same question: A] / [CO4] Group A: 1 = Male, 2 = Female, 3 = Other Group B: 0 = Male, 1 = Female, 2 = Other What problem will this cause when combining both sets of data, and how will you solve it?

The two groups used different coding systems for the same variable ("Gender"). As a result, when combining both datasets, the codes will not match correctly — ***1 mark***

Example, *"1" means Male in one set but Female in the other.*
This inconsistency will lead to errors in analysis and misleading results, since the data cannot be compared or merged accurately. – ***1 mark***

Solution*: 1 marks*
Before combining the data, the researcher should recode the variable so that both groups use the same coding system.
For instance, convert all responses to: 1 = Male, 2 = Female, 3 = Other

Part B

Short Essay

6a. Why is it necessary for researchers to handle data with transparency and honesty? [U] / [CO1]

Give marks for any valid points

It is necessary for researchers to handle data with transparency and honesty to maintain the credibility, accuracy, and ethical integrity of their work.

- **Ensures Trustworthiness:** Honest data handling builds trust among readers, participants, and the research community.
- **Prevents Misleading Results:** Transparency helps avoid data manipulation or fabrication, ensuring valid and reliable findings.
- **Supports Ethical Standards:** Researchers have a moral duty to report data truthfully and allow others to verify or replicate their studies.

6b. Differentiate between pure research and applied research using one example from daily life. [U] / [CO1]

Meaning and purpose 3 marks/ example 2 mark

Basis	Pure Research	Applied Research
Meaning	Aims to expand knowledge and understand underlying principles without immediate practical use.	Aims to solve specific, practical problems using existing knowledge.
Purpose	To develop theories or concepts.	To find solutions or improvements for real-world issues.
Example	Studying how the human brain processes stress to understand its mechanism.	Developing a stress management program for college students based on those findings.

7a. Discuss two practical problems that may arise if the layout of a research report is not consistent. [U] / [CO4]

A well-structured research report follows a systematic layout that helps readers clearly understand the study's purpose, process, and findings. Inconsistent layout causes **confusion** and **loss of credibility**, affecting both the readability and acceptance of the research report.

7b. Explain the importance of research reports in business decision[A] / [CO4]

Provide 3marks for similar points and 2 marks for examples.

Research reports are essential tools for ***analyzing facts, identifying opportunities, and guiding sound business decisions*** through evidence-based insights.

Factual information *Example:* A market research report showing customer preferences helps a company decide which new product features to introduce.

identifying opportunities *Example:* A report on regional demand helps a business decide where to open a new branch.

8a. Describe the steps in formulating research objectives with reference to a study on online shopping behavior. [U] / [CO2]

Provide 2.5 mark for steps. Full marks for explanation.

First, identify the research problem — for example, we want to study why people prefer online shopping. Second, review related studies to understand what other researchers have

found about online shopping behavior. Third, define the purpose of the study clearly, such as finding out factors that influence customers to shop online. Fourth, frame specific and measurable objectives like "to study the effect of price discounts on online purchase decisions" or "to analyze how trust and website design affect customer satisfaction." Finally, check that the objectives are realistic and related to the main problem so that they guide the data ollection and analysis.These steps help the researcher stay focused and ensure that the study on online shopping behavior is clear, meaningful, and achievable.

8b. Explain the importance of sampling design in a research study. [U] / [CO2]

2.5 marks for points, full mark for explanation.

Saves Time and Cost; Ensures Accuracy; Reduces Bias; Improves Efficiency; Enables Generalization; Enhances Validity.

Part C

Long Essay

9a. A company introduces a work-from-home policy and wants to evaluate its impact on employee performance. Suggest best practices a researcher should follow to frame effective research questions for this study. Also explain in detail criteria for good research question. [U] / [CO2]

Introduction and conclusion- 4 marks/ best practices – 8 marks/Criteria- 8 marks

Best Practices to Frame Effective Research Questions

1. **Understand the purpose** – Identify what the company wants to find out through the study.
2. **Review related literature** – Study previous research to find knowledge gaps.
3. **Be clear and specific** – Frame focused and easy-to-understand questions.
4. **Focus on measurable aspects** – Ensure questions can be answered with data.
5. **Ensure relevance** – Align questions with the company's goals and objectives.
6. **Avoid bias** – Frame neutral and unbiased questions.
7. **Check feasibility** – Make sure questions are practical and realistic to answer.
8. **Seek feedback** – Get expert or peer review before finalizing the questions.

Criteria for a Good Research Question

1. **Clarity:** The question must be clear and easy to understand, leaving no confusion about its meaning.
2. **Specificity:**It should focus on a particular issue or variable rather than being too broad.
3. **Relevance:** The question must be connected to the research problem and objectives of the study.
4. **Feasibility:** It should be practical to answer within the limits of time, cost, and data availability.
5. **Measurability:**The question should be capable of being answered through data collection, observation, or analysis.
6. **Originality:** It should add some new insight or understanding to the existing knowledge about work-from-home and performance.

7. **Ethical Soundness:** The question should not invade privacy or cause harm to participants. OR

The **FINER Criteria** is a useful guideline for developing a good research question. It stands for **Feasible, Interesting, Novel, Ethical, and Relevant.**

9b. Discuss the research design in descriptive and diagnostic studies. Explain the steps involved, including formulating objectives, designing data collection methods, selecting samples, collecting, processing, analyzing data, and reporting findings, and highlight measures to reduce bias and maximize reliability. [U] / [CO2] –

Introduction and conclusion – 5 marks/Explanation for both research design – 5 marks/ Aspects – 10 marks

Aspect	Explanation
Type of Study	**Descriptive Study:** Describes characteristics, behavior, or trends of a group or situation.**Diagnostic Study:** Describes and also identifies causes or reasons for a problem.
Example	Descriptive – Studying online shopping behavior.Diagnostic – Finding reasons for employee performance decline after work-from-home.
1.Formulating Objectives	Define what the study aims to achieve, e.g., "To study customer satisfaction" or "To identify factors affecting motivation."
2. Designing Data Collection Methods	Choose suitable tools like questionnaires, surveys, or interviews to collect relevant data.
3. Selecting Samples	Use correct sampling techniques (random, stratified, cluster) to represent the population accurately.
4. Collecting Data	Gather data systematically with clear instructions to respondents to ensure accuracy.
5. Processing Data	Edit, code, and classify data to remove errors and make it ready for analysis.
6. Analyzing Data	Apply statistical tools to interpret results and find patterns or causes related to objectives.
7. Reporting Findings	Present results using charts, graphs, and tables. Summarize conclusions and make recommendations.
Measures to Reduce Bias	- Use random sampling- Avoid leading questions- Maintain uniform procedures- Conduct pilot tests
Ways to Maximize Reliability	- Use standardized tools- Ensure consistency in data collection- Double-check data accuracy
Conclusion	A well-structured research design ensures systematic data collection, accurate analysis, reduced bias, and reliable findings.

10a. A researcher wants to study the effect of advertisements on sales of a product. Identify the variables, suggest suitable scales of measurement, and explain which scaling technique you would apply to measure advertisement effectiveness. [A] / [CO3]

i) Identification of Variables (6 Marks)

- Independent Variable (Cause): Advertisement (the factor whose effect is being studied). This could include factors like type of advertisement, frequency, duration, media used, message content, etc.
- Dependent Variable (Effect): Sales of the product (the outcome influenced by advertisements; This measures how much the product is sold after the advertising activity.)
- Possible Control Variables: Price, distribution, season, brand image, etc. (optional mention for higher accuracy).

ii) Scales of Measurement (Explaining the various scales (5 marks) and what the student plans to use in this study (2 marks))

Type of Scale	Description	How it is used in this study (full marks for any one example)
Nominal Scale	Used for classifying data without any order or numerical value. It simply labels variables.	Used to classify types of advertisements (e.g., TV, print, online, social media). Each type is a separate category with no ranking implied.
Ordinal Scale	Indicates rank order among variables but does not show equal intervals between ranks.	Can be used to rank advertisement preference or effectiveness level (e.g., very effective, effective, neutral, less effective, not effective).
Interval Scale	Measures variables where differences between values are meaningful, but there is no true zero point.	Can be used in attitude scales like the Likert scale to measure agreement with statements about advertisement effectiveness (e.g., from Strongly Agree to Strongly Disagree).
Ratio Scale	Has all the properties of interval scale, and also has a true zero point, allowing comparison of absolute magnitudes.	Used to measure sales figures (in rupees or units sold) or advertisement frequency (number of ads shown). These have meaningful zero points.

iii) Scaling Technique to Measure Advertisement Effectiveness (Explaining various scaling techniques (5 marks) and what the student plans to use in this study, with sample questions (3 marks))

Scaling Technique	Description	Type of Scale	Example (Application) (full marks for any one points)	Use in Study (full marks for any one points)
Likert Scale	Respondents indicate their level of agreement or disagreement with given statements about the advertisement.	Interval Scale	"The advertisement is convincing." (1 = Strongly Disagree → 5 = Strongly Agree)	Measures consumers' attitude and perception toward the advertisement. Scores are summed or averaged for overall effectiveness.

Semantic Differential Scale	Uses pairs of opposite adjectives to measure the emotional response toward the advertisement.	Interval Scale	Interesting ☐ ☐ ☐ ☐ ☐ Boring	Measures emotional tone or brand image created by the advertisement.
Rating Scale	Respondents assign a numerical value to indicate how effective they find the advertisement.	Ordinal Scale	"Rate the effectiveness of this advertisement on a scale of 1 to 10."	Provides a simple quantitative measure of perceived effectiveness, suitable for comparison.

10b. Define Survey and enlist the different Types of Survey? [K] / [CO3]

Introduction and conclusion – 5 marks/Definition or meaning- 5 marks/ difference-10 marks

Type	**Description**
1. Personal Interview Survey	Data is collected through face-to-face interaction between interviewer and respondent.
2. Telephone Survey	Information is gathered by calling respondents over the phone.
3. Mail Survey	Questionnaires are sent to respondents by post, and they fill and return them.
4. Online/Email Survey	Conducted through internet platforms or email links for quick responses.
5. Observation Survey	Researcher observes and records behavior without direct questioning.
6. Group or Panel Survey	Data is collected from a fixed group of respondents (panel) over time to track changes.

Prepared by,

Dr. Anju E N,

Sree Narayana Arts and Science College, Kumarakom

8098851065